PRE-ALGEBRA (8TH) · UNIT 2: EQUATIONS AND EXPRESSIONS

Phone Plan Showdown

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students act as consultants at a phone store. They write cost expressions, solve equations to find break-even points, and classify equations by their number of solutions.

The plans make every equation meaningful: a solution is the amount of data where two plans cost the same.

Standards Addressed

- 8.EE.C.7a One, no, or infinitely many solutions
- 8.EE.C.7b Solve linear equations with rational coefficients, distributive property and like terms
- 7.EE.A.1 Simplify expressions

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with three real plan ads. Part 1: writing and simplifying cost expressions.
Day 2	Part 2: solving equations from bills and break-even points.
Day 3	Part 3: special cases and recommendations; build the customer guide.

Materials

- Student handout with the plan cards
- Calculators
- Graph paper (optional for graphing the plans)
- Poster or slide template for the customer guide

How the Three Versions Differ

Level	What changes
CP2	Two-step equations with starters and one break-even equation.
CP1	Distributive property, variables on both sides, solution types, and ranges where a plan is cheapest.
Honors	Tiered pricing, average cost, parameter questions, and a designed plan with an algebraic proof.

Watch For (Common Misconceptions)

- Distributing to only the first term in $3(g - 2)$.
- Moving a term across the equals sign without changing its sign.
- Writing "x = 0" when an equation has no solution.

Extension Ideas

- Graph all plans on one coordinate plane and mark the break-even points.
- Research two real plans and repeat the analysis.
- Add a one-time phone cost and compare plans over two years.

Answer Key – CP2

Part 1: Write the Costs

#	Task	Answer
1	Write an expression for the monthly cost of Plan A and of Plan B.	Plan A: 15 + 5g. Plan B: 30 + 2g.
2	Find the cost of each plan for a customer who uses 4 GB.	Plan A: 15 + 5(4) = $35. Plan B: 30 + 2(4) = $38.
3	Simplify 3(5g + 2) − 4g.	15g + 6 − 4g = 11g + 6.

Part 2: Solve for the Data

#	Task	Answer
4	A Plan A bill was $50. How many GB were used? Solve 15 + 5g = 50.	5g = 35, so g = 7 GB.
5	A Plan B bill was $46. How many GB were used?	30 + 2g = 46 → 2g = 16 → g = 8 GB.
6	Solve 2(x + 4) = 18.	2x + 8 = 18 → 2x = 10 → x = 5.

Part 3: Which Plan Wins?

#	Task	Answer
7	For how many GB do Plans A and B cost the same? Solve 15 + 5g = 30 + 2g. What is the cost?	3g = 15, so g = 5 GB. Both cost $40.
8	Which plan is cheaper for 3 GB? For 8 GB?	3 GB: A = $30, B = $36, so A. 8 GB: A = $55, B = $46, so B.
9	At how many GB does Plan B cost the same as Plan C ($60)?	30 + 2g = 60 → g = 15 GB.

Reflection

#	Prompt	Look for
R1	What does the solution of 15 + 5g = 30 + 2g mean for a customer?	At 5 GB both plans cost $40; below 5 GB Plan A is cheaper and above 5 GB Plan B is cheaper.
R2	Why do you do the same thing to both sides of an equation?	To keep the two sides equal, so the solution does not change.

Answer Key – CP1

Part 1: Plans as Expressions

#	Task	Answer
1	Write expressions for the monthly cost of Plans A, B, and C.	A: 15 + 5g. B: 30 + 2g. C: 60.
2	A family has 3 lines on Plan A, each using g GB, plus a $10 family fee. Write and simplify the total cost.	3(15 + 5g) + 10 = 45 + 15g + 10 = 15g + 55.
3	A customer uses g GB in month 1 and 3 more GB in month 2. Write and simplify the two-month cost on Plan A.	(15 + 5g) + (15 + 5(g + 3)) = 10g + 45.

Part 2: Solving Equations

#	Task	Answer
4	Solve 15 + 5g = 30 + 2g and explain what the solution means.	g = 5. At 5 GB both plans cost $40.
5	Solve 3(g − 2) + 4 = 2g + 7.	3g − 6 + 4 = 2g + 7 → 3g − 2 = 2g + 7 → g = 9.
6	The family's Plan A bill was $160. How many GB did each line use?	15g + 55 = 160 → 15g = 105 → g = 7 GB.

Part 3: Special Cases & Recommendations

#	Task	Answer
7	How many solutions does each equation have? Explain. (a) 2(5g + 15) = 10g + 30 (b) 5g + 15 = 5g + 20	(a) Infinitely many: both sides simplify to 10g + 30. (b) No solution: 15 ≠ 20 after subtracting 5g.
8	For which amounts of data is Plan B the cheapest of all three plans?	B < A when g > 5; B < C when g < 15. So Plan B is cheapest for 5 < g < 15 GB.
9	A customer averages 12 GB each month. Which plan do you recommend? Show the costs.	A: 15 + 60 = $75. B: 30 + 24 = $54. C: $60. Recommend Plan B.

Reflection

#	Prompt	Look for
R1	What does it mean in context when an equation has no solution?	The two plans never cost the same; one is always more expensive by the same amount.
R2	How did combining like terms help you compare plans?	Simplifying each side shows the total rate and fixed cost, so the equation is easier to solve and interpret.

Answer Key – Honors

Part 1: Tiered Plans

#	Task	Answer
1	Write a simplified expression for the cost of Plan D when g > 5.	10 + 6(5) + 3(g − 5) = 10 + 30 + 3g − 15 = 3g + 25.
2	Write an expression for the average cost per GB on Plan B.	$\frac{30+2g}{g}$ dollars per GB.
3	Write and simplify an expression for how much more Plan A costs than Plan B over a year for g GB each month.	12[(15 + 5g) − (30 + 2g)] = 12(3g − 15) = 36g − 180.

Part 2: Equation Analysis

#	Task	Answer
4	When do Plans D and B cost the same? Check that your answer fits the condition g ≥ 5.	3g + 25 = 30 + 2g → g = 5. At 5 GB both cost $40 (D: 10 + 30 = 40), and g = 5 is at the boundary where the formula applies.
5	Solve $\frac{1}{3}(6g - 9) + 2 = 2(g - 1) + 1$. What kind of solution set is it?	2g − 3 + 2 = 2g − 2 + 1 → 2g − 1 = 2g − 1: infinitely many solutions.
6	For what value of k does 4g + k = 4g + 12 have infinitely many solutions? For which values does it have no solution?	k = 12 gives infinitely many solutions; any k ≠ 12 gives no solution.

Part 3: Design a Plan

#	Task	Answer
7	For how many GB is the average cost per GB on Plan B exactly $5?	$\frac{30+2g}{g} = 5$ → 30 + 2g = 5g → g = 10 GB.
8	Design a plan that costs more than Plan A at 2 GB but less than Plan B from 8 GB to 12 GB. Show the costs at 2, 8, and 12 GB.	Answers vary. Example: $28 + $2 per GB. At 2 GB: $32 > A ($25). At 8 GB: $44 < B ($46). At 12 GB: $52 < B ($54).
9	Use algebra to prove whether your plan is cheaper than Plan B for every amount of data.	Example: 28 + 2g < 30 + 2g is true for all g because 28 < 30, so the plan is always $2 cheaper.

Reflection

#	Prompt	Look for
R1	Why does the coefficient of g decide whether two plans can ever cost the same?	If the rates (coefficients) are equal, the costs differ by a constant forever (no solution or identical); different rates always meet once.

#	Prompt	Look for
R2	How is the average cost per GB different from the price per GB?	Average cost spreads the monthly fee over the data used, so it drops as data grows; the price per GB is only the rate.